

Mehul Agarwal

Software Engineer | Data Engineering | Backend Development

Noida, India | +91 8979011405 | mehulagarwal1313@gmail.com | linkedin.com/in/mehul-agarwal | github.com/mehulagarwal1313

SUMMARY

Computer Science graduate (2026) with hands-on experience building scalable ETL/ELT data pipelines, distributed backend systems, and ML-integrated services that process large datasets. Proficient in Python, SQL, Apache Spark, and cloud platforms (Microsoft Azure, AWS). Skilled at optimizing performance, improving data quality, and collaborating in Agile teams. Strong computer science fundamentals with 700+ data structures and algorithms problems solved.

TECHNICAL SKILLS

Languages: Python, TypeScript, C++, SQL, JavaScript

Data Engineering: ETL/ELT Pipelines, Apache Spark, Microsoft Fabric, Snowflake, Data Modeling, Data Quality, Data Validation, Pandas, NumPy

AI/ML and GenAI: LLMs, LangChain, LangGraph, RAG, Model Context Protocol (MCP), TensorFlow, Vector Databases, Prompt Engineering

Cloud and DevOps: Microsoft Azure, AWS (S3), Docker, CI/CD, Git, GitHub

Web and Backend: FastAPI, Node.js, Express.js, React.js, REST API Design, WebSockets, Server-Sent Events

Databases: PostgreSQL, MySQL, MongoDB, Qdrant

CS Fundamentals: Data Structures and Algorithms, System Design, Distributed Systems, Object-Oriented Programming

WORK EXPERIENCE

Navikenz

Mar 2026 – Jun 2026

Software Engineer Intern, Data Engineering and Backend

Noida, India

- Engineered ETL ingestion pipelines on Microsoft Fabric, consolidating data from **3 heterogeneous sources** (Snowflake, Amazon S3, SharePoint) into unified target databases, cutting manual data-transfer effort by **60%**.
- Designed source-to-target mappings and Apache Spark transformations across **5+ schemas**, applying validation and data-quality checks that reduced ingestion errors by **30%** and ensured production-grade reliability.
- Partnered with cross-functional engineers and product stakeholders in an Agile environment to diagnose pipeline failures, resolving **100%** of reported data-flow incidents within agreed SLAs.

Pristine Business Solutions

May 2025 – Jul 2025

Software Developer Intern, Enterprise Systems and Data Validation

Noida, India

- Implemented automated data-validation pipelines that improved workflow reliability by **35%** across enterprise production systems.
- Optimized complex SQL queries through index tuning and query restructuring, reducing average execution time by **40%** on large datasets.
- Developed anomaly-detection monitoring tools and automated reporting workflows, increasing data accuracy and accelerating operational decision-making.

PROJECTS

NaviformatIQ – AI Multi-Agent Document Reformatting Platform

GitHub Live

- Architected an AI-driven, multi-agent platform with **3 domain-agnostic workflows** (document generation, style transfer, compliance checking) using LangGraph and Azure OpenAI; maps draft content into a template structure via RAG and surfaces a side-by-side diff for human review.
- Built a FastAPI backend orchestrating multi-agent pipelines with Neon Postgres, Qdrant vector search, and Server-Sent Events for real-time streaming of agent progress.
- Designed a React (Vite + Tailwind) wizard with an interactive human-in-the-loop review flow and a context-aware chat agent; exports final reports to .docx and .pdf.
- Tech Stack:** Python, FastAPI, LangGraph, RAG, React, Docker, Qdrant, PostgreSQL, Azure OpenAI

Sovaria – Authority-Driven Backend-as-a-Service Platform

GitHub Live

- Constructed a distributed Backend-as-a-Service spanning **11 specialized engines** (identity, policy, NoSQL/SQL storage, temporal versioning, realtime, simulation, functions, storage, billing) behind a single mandatory Identity → Policy → Execution pipeline.
- Implemented dual data engines – a NoSQL store (LSM-tree, Bloom filters, compaction) and an MVCC SQL engine with a query planner and cross-engine JOINS – with policy compiled to a decision tree for sub-millisecond evaluation.
- Delivered low-latency WebSocket realtime (5–20 ms from commit), an append-only version graph with zero-copy snapshots, and identical Python and TypeScript SDKs with a CLI supporting dry-run simulation for CI/CD.
- Tech Stack:** Python, TypeScript, Distributed Systems, NoSQL and SQL Engines, WebSockets, REST API, Git

Code Similarity and Plagiarism Detection System

GitHub Live

- Developed a plagiarism-detection system using AST parsing, tokenization, and control-flow analysis to identify structural code similarities across large code datasets.
- Streamlined multi-file processing pipelines to compute pairwise similarity scores efficiently, with an interactive dashboard that ranks and highlights suspicious code pairs.
- Tech Stack:** Python, Flask, Algorithms, SQL

EDUCATION

Jaypee Institute of Information Technology

2022 – 2026

Bachelor of Technology, Computer Science and Engineering

Noida, India

- CGPA: 8.2 / 10

ACHIEVEMENTS

Solved **700+** data structures and algorithms problems on LeetCode and Code360, focused on optimization and scalable solutions.